Final Review
Fall Prevention
Exoskeleton Clinic

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Project Scope

What are we doing here?

Problem Statement

- Falls are a leading cause of injuries among the elderly and individuals with mobility impairments.
- Exoskeleton technology has the potential to mitigate these risks by providing real-time support during gait perturbations.
- Our project focuses on developing a hip and knee exoskeleton capable of detecting and responding to these perturbations, with the goal of improving stability and mobility.

Problem Statement

- Last semester, preliminary bench and human subject test results demonstrated the the capabilities of the current exoskeleton design.
- This semester's goal is to improve upon last year's exoskeleton design and test results with a focus on increasing functionality.

02

Deliverables

What have we done?

Deliverables

Deliverable	Description
Add Hip Flexion Capabilities	— Design hip flexion function — Determine strength of design
Thigh Brace Improvements	— Added more straps to add more sturdiness of thigh brace when put all together with hip components. — Added different support components to add force and prevent unwanted bending from the brace.
Abduction Cable	— Redesign abduction cable holder to be more durable and secure to hip brace — Redesign method of connecting cable to moment arm so it can be easily detached and reattached

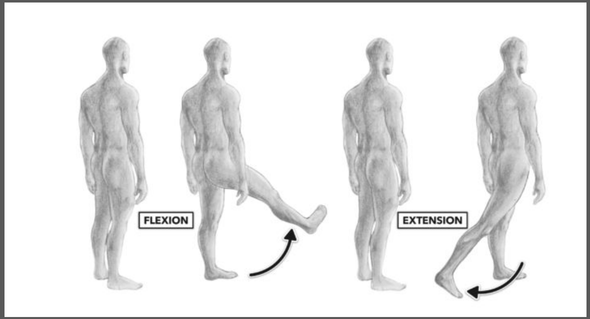
Deliverables

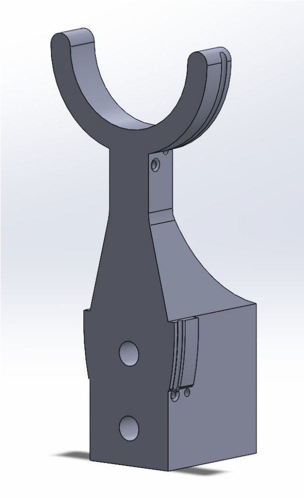
Deliverable	Description
IMU Hardware	— Design of custom IMU harness + PCB for data collection
Control Board PCB	— Custom Control Board PCB for overall system control (solenoids, relays, pressure transducers) and safety features (temperature sensing, current sensing) for emergency shutoff
Exoskeleton Test Bench	- Test Bench to test hardware functionality: Solenoids, Solid State Relay, IMU System to Control Board Communication, Power Systems

Hip Flexion Addition

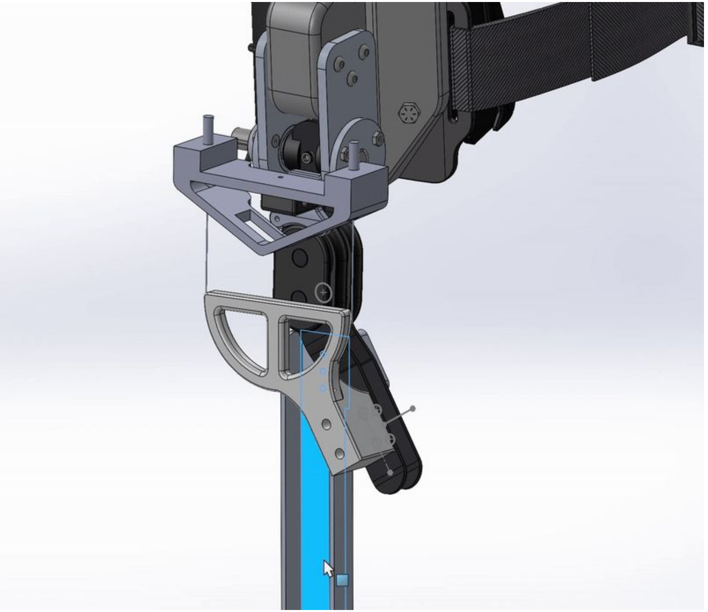
While the leg is currently able to abduct, we are unable to move the leg forward or backward.

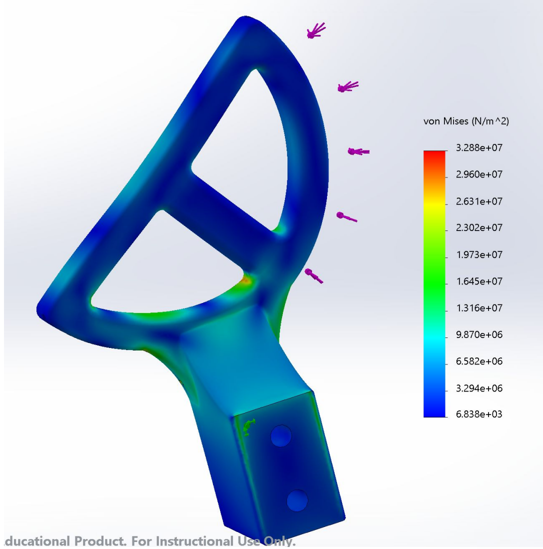
- Goal: 60° motion forward, 30° backward

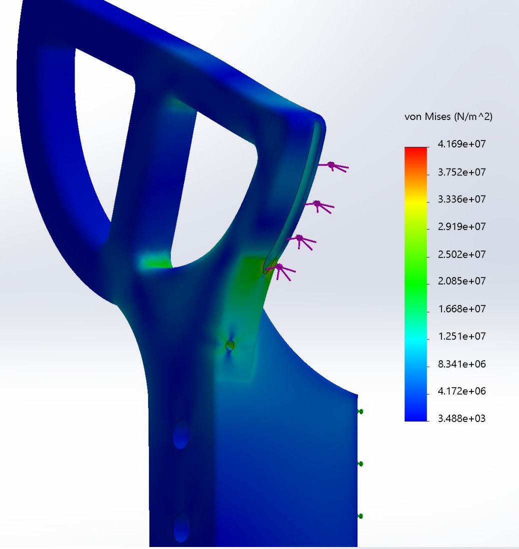




Hip Flexion - Final Design

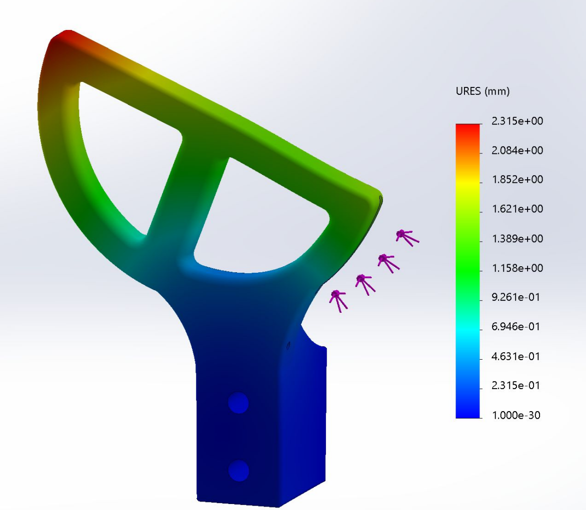






Stress Analysis:
- ABS Material
- 500N force along cable channel

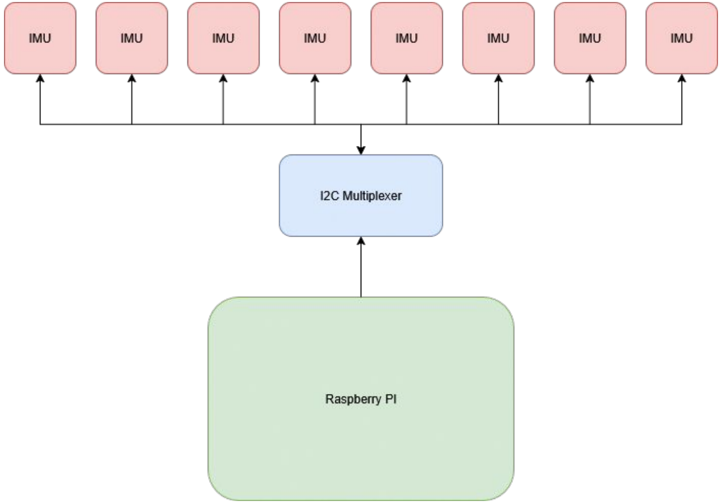
Max stress: 4.2 MPa
Max disp: 2.3 mm



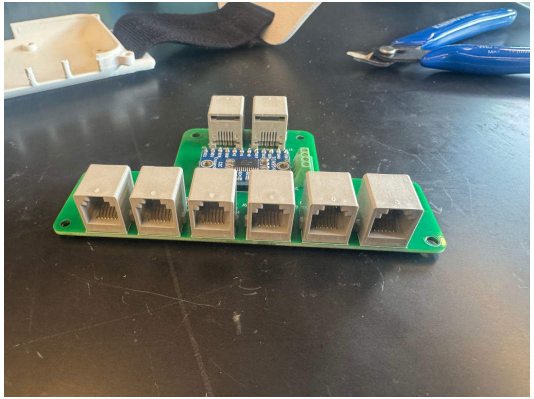
IMU System

Key Features:

- Used for gate detection and gate DAQ
- Manage 8 IMUs simultaneously via I²C
- Modular RJ12 connectors for each IMU
- Provide appropriate signal filtering and minimize dropout at greater wire lengths
- Interface with a Raspberry Pi 4
- Compact, wearable form factor



IMU System fully assembled. Strap fastening points on either end for wearability during data acquisition.

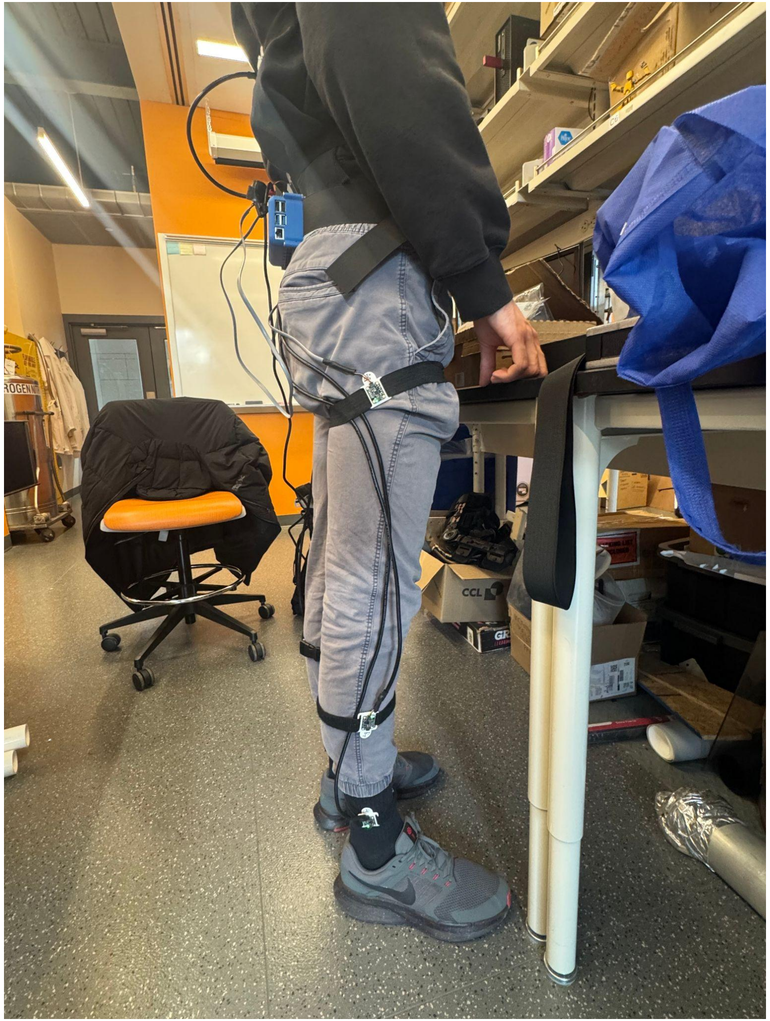


IMU PCB fully populated. 8 RJ12 Connectors for modularity

IMU System DAQ

Key Features:

- 8 IMUs for both legs
- Upper Back, Lower Back, Thigh, Calf, Ankle
- IMUs are worn with custom fitted straps and in a 3D printed holder



IMU System DAQ

DAQ Process:

- Accessed through SSH
- Automatically logs data in Quaternions
- After data collection is halted, a server is created for file access and downloading

IMU Straps worn during DAQ

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	Timeleft_thigh_wleft_thigh_xleft_thigh_yleft_thigh_zright_thigh_wright_thigh_xright_thigh_yright_thigh_zleft_calf_wleft_calf_xleft_calf_yleft_calf_zright_calf_wright_calf_xright_calf_yright_calf_zleft_ankle_wleft_ankle_xleft_ankle_yleft_ankle_zright_ankle_wright_ankle_xright_ankle_yright_ankle_zlower_back_																									
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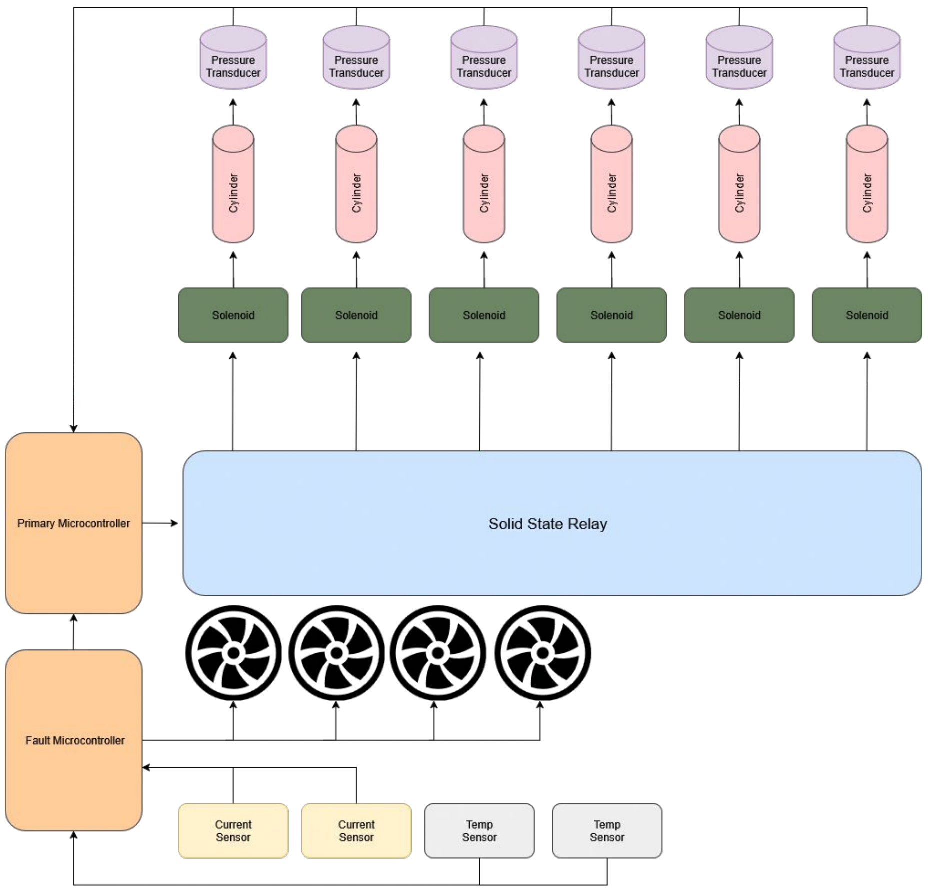
IMU Data Set in CSV Format

Exoskeleton Control System

Key Features:

- Drive solenoids through relay control
- Control cooling fans
- On-board current sensing for input current
- Interface with Adafruit current sensor boards and pressure transducer modules (COTS)
- Distribute power safely and reliably
- Fit within wearable exoskeleton chassis
- Include both a primary microcontroller (basic functions) and a secondary microcontroller (fault detection)



Exoskeleton Control Board

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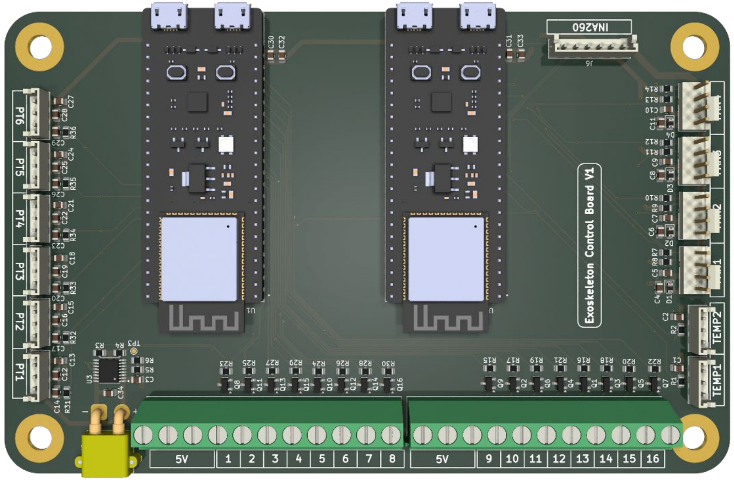
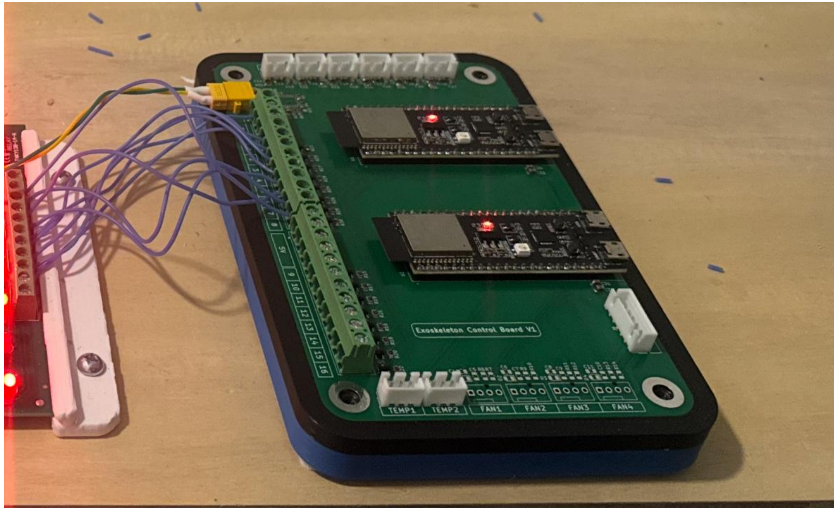
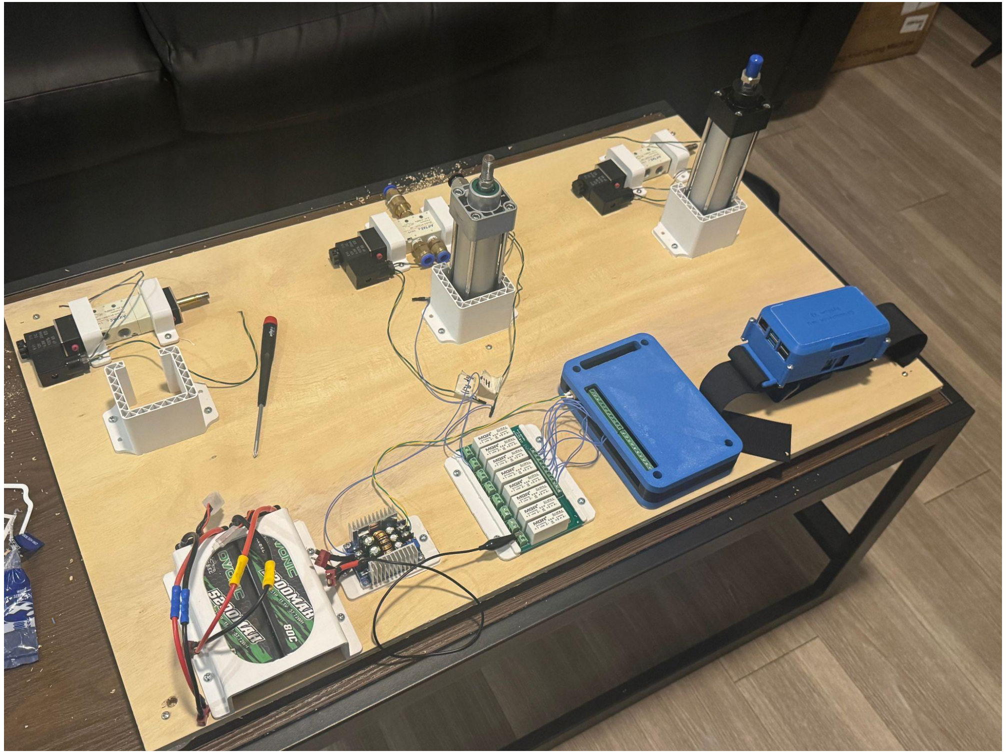




Exoskeleton Test Bench

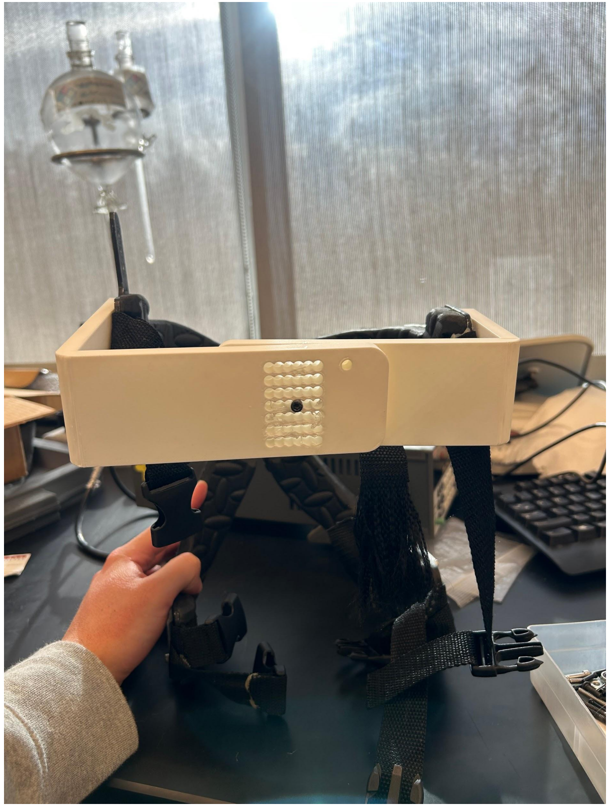


Thigh Brace

- Original Problem
    - The brace was too loose, uncomfortable, and there was a large bending/twisting force that needed to be addressed.
- Two extra straps were added to go straight across the brace.
- This improved the sturdiness of the brace, while also preventing the straps from loosening from moving.

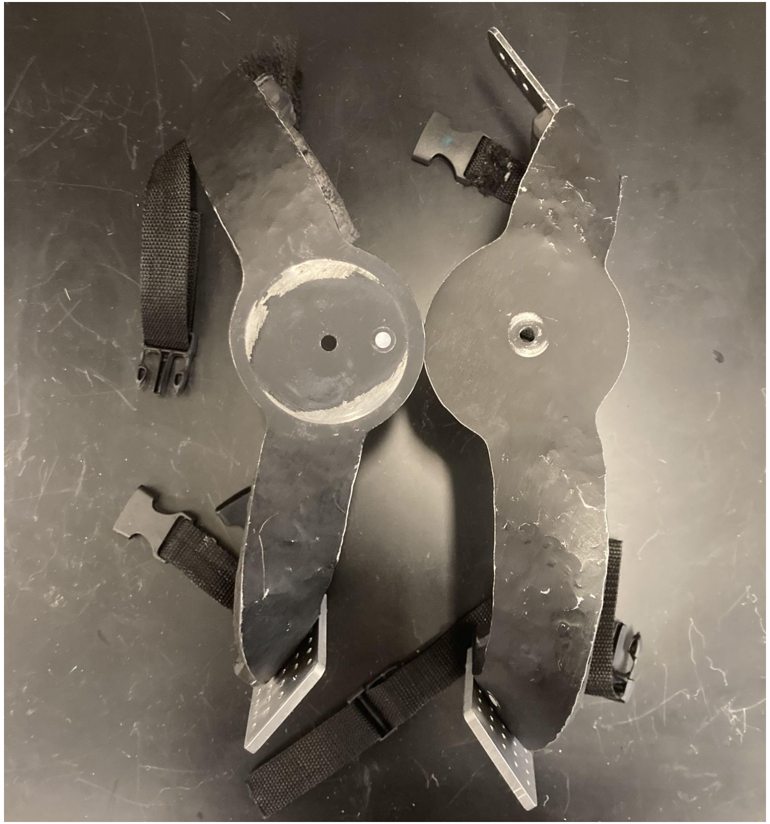
Thigh Brace

- Developed Bracket supports to add more force to prevent bending in the brace
- The pattern in the middle allows brackets to move smoothly when adjusting the brace for comfortability.



Thigh Brace

- Halves tend to twist



Thigh Brace

- Circular spacer with designed holes made and used to drill holes in the brace
- Designed for a hole to line up every 3° of motion and to fit
- The purpose of the holes is to add screws and prevent the current twisting motion caused by brace movement

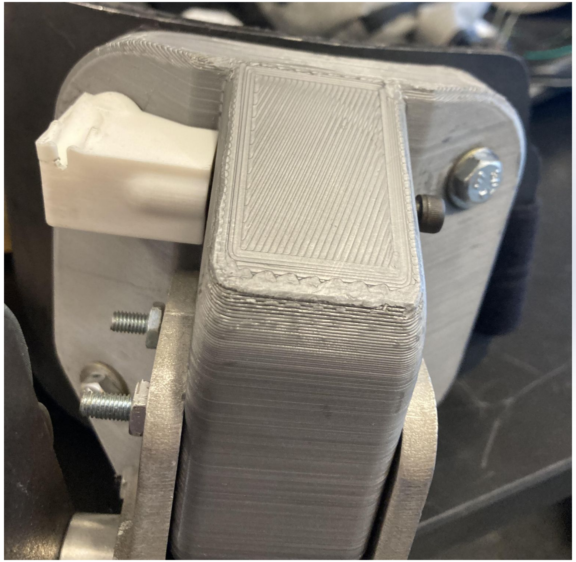
Abduction Cable

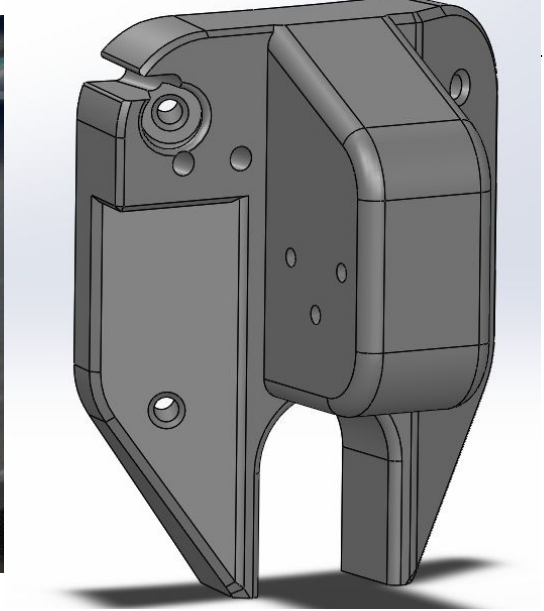
Problems

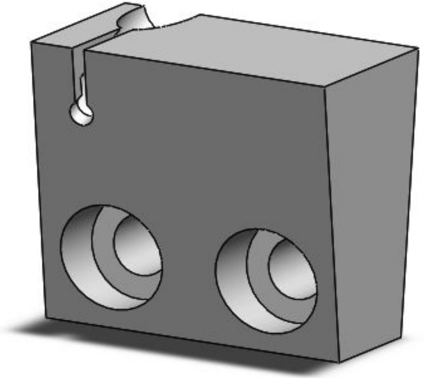
- Steel cable wears down holder
- Sleeve can easily pop out of place
- Structurally weak

Possible Solutions

- Increase thickness of the front wall
- Change material around the hole to TPU
- Add screws to front
- Secure holder with hip brace bolt



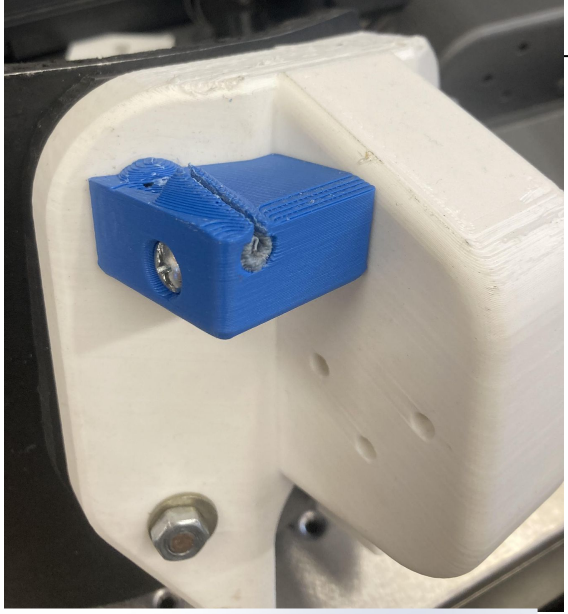


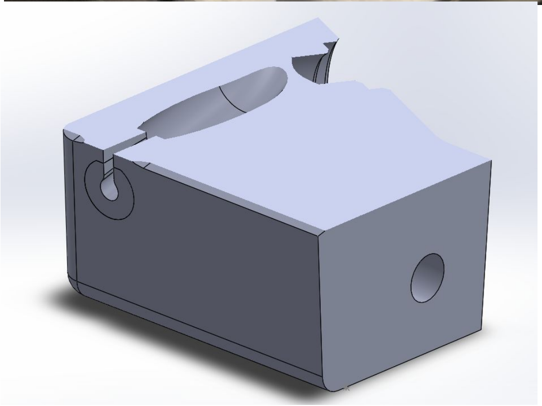


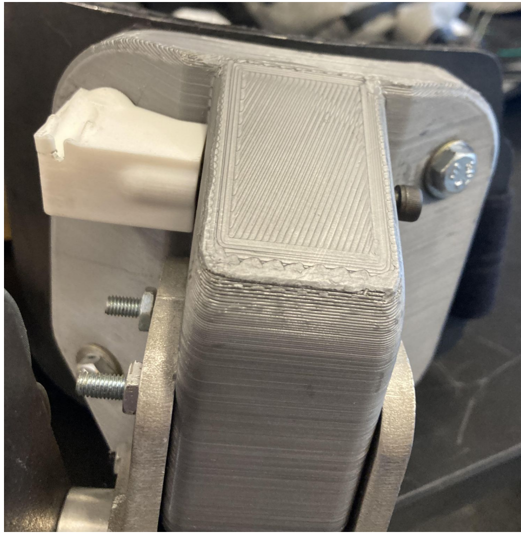
Abduction Cable

Chosen Solution

- Decided to 3D print part
- Increased hole wall thickness
- Added screw with heatset insert
- Made layers around hole out of strengthened TPU to decrease wear







03

Future Work

Where are we going?

Future Work

Deliverable	Description
Hip Flexion Addition	— Continue developing a sturdy cable holder — Begin rapid prototyping and testing
Thigh Brace	— Test and compare how much the improvements increase the strength
Abduction Cable	— Test and see if the steel cable still wears it down
IMU System V2	- More reliable DAQ, upgraded IMU's, switch to UART

Thank you!

And a special thanks to Dr. Trkov and Vaibhavsingh Varma our advisors!

Any Questions?

04

Appendix

Standards and Regulations

Standard	Description
IEC 60601-1	Safe power distribution, grounding, and insulation
CISPR 11, EN 61000	Ensure no interference with other devices
ISO/IEC 14908	Data Exchange
ISO 5725/ ISO 6789	Maintain sensor precision/calibration
ISO 14791	Risk management and safety evaluations
ISO 13850	Emergency stop/disconnect
Pilz 8.3.3, 8.5.2	Installation of pneumatic linear actuators

*This is only some of the many standards we follow for this project

Muscle Activation During Slip & Slip Recovery

OpenSim

A free software program that allows for the modeling of mechanical and biological structures

Goal

To use data collected from IMU’s to create a virtual model of the musculoskeletal system, during slips & recovery.

How It Works

Reading Data

The IMU data is read and converted in OpenSim files

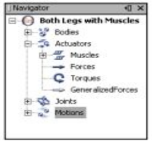
Post Processing

Rotate lab coordinates to match OpenSim coordinate system. Compute force center of

Results

- Perform inverse kinematics
- Perform inverse dynamics
- Use that to calculate

OpenSim GUI

GUI Components	
Title	Screenshot
Toolbar	
Motion Text Box	
Motion Slider / Video Controls	
View Window	
Navigator Window	
Coordinates Window	
Muscle Editor